

APPENDIX

SECTION 2.1

Figure 2.1.1 - ER MODEL

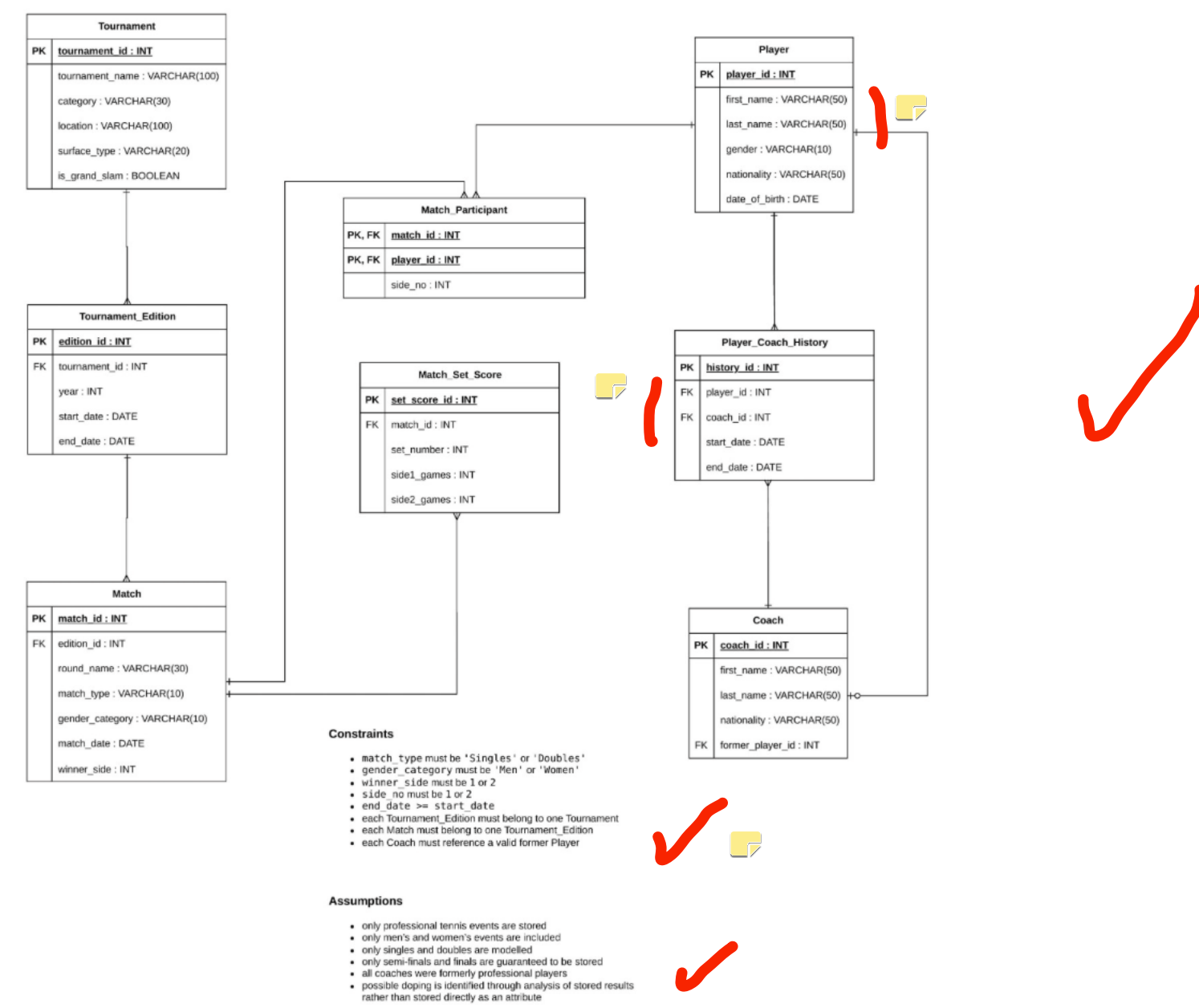


Figure 2.1.2.a: Full entity specification table showing entity purposes, attributes, primary keys, foreign keys and important constraints.

Entity	Purpose	Attributes and domains	Primary key	Foreign key(s)	Important constraints / notes
Tournament	Stores the general details of each tennis tournament.	<u>tournament_id</u> INT, <u>tournament_name</u> VARCHAR(100), category VARCHAR(30), <u>location</u> VARCHAR(100), <u>surface_type</u> VARCHAR(20), <u>is_grand_slam</u> BOOLEAN	<u>tournament_id</u>	None	Stores tournament-level details that do not change for each yearly edition, such as name, location, surface type and whether it is a Grand Slam.
Tournament Edition	Stores a specific yearly edition of a tournament.	<u>edition_id</u> INT, <u>tournament_id</u> INT, <u>year</u> INT, <u>start_date</u> DATE, <u>end_date</u> DATE	<u>edition_id</u>	<u>tournament_id</u> → Tournament(<u>tournament_id</u>)	Each tournament edition must belong to one tournament. <u>end_date</u> should be greater than or equal to <u>start_date</u> . This separates the repeating yearly event from the general tournament information.
Match	Stores details of each tennis match in a tournament edition.	<u>match_id</u> INT, <u>edition_id</u> INT, <u>round_name</u> VARCHAR(30), <u>match_type</u> VARCHAR(10), <u>gender_category</u> VARCHAR(10), <u>match_date</u> DATE, <u>winner_side</u> INT	<u>match_id</u>	<u>edition_id</u> → TournamentEdition(<u>edition_id</u>)	Each match must belong to one tournament edition. <u>match_type</u> must be either 'Singles' or 'Doubles'. <u>gender_category</u> must be either 'Men' or 'Women'. <u>winner_side</u> must be 1 or 2.
Player	Stores details of each professional tennis player.	<u>player_id</u> INT, <u>first_name</u> VARCHAR(50), <u>last_name</u> VARCHAR(50), <u>gender</u> VARCHAR(10), <u>nationality</u> VARCHAR(50), <u>date_of_birth</u> DATE	<u>player_id</u>	None	Player details are stored once and then reused through relationships with matches, coach history and coaches. This avoids repeating player information in match records.
Match Participant	Links players to matches and records which side each player played on.	<u>match_id</u> INT, <u>player_id</u> INT, <u>side_no</u> INT	Composite primary key: <u>match_id</u> , <u>player_id</u>	<u>match_id</u> → Match(<u>match_id</u>); <u>player_id</u> → Player(<u>player_id</u>)	This resolves the many-to-many relationship between Player and Match . <u>side_no</u> must be 1 or 2. This structure allows both singles and doubles matches because multiple players can be assigned to each side.
Match Set Score	Stores the score for each set in a match.	<u>set_score_id</u> INT, <u>match_id</u> INT, <u>set_number</u> INT, <u>side1_games</u> INT, <u>side2_games</u> INT	<u>set_score_id</u>	<u>match_id</u> → Match(<u>match_id</u>)	Each set score belongs to one match. <u>set_number</u> identifies which set is being recorded. <u>side1_games</u> and <u>side2_games</u> store the games won by each side in that set.
Coach	Stores details of each coach.	<u>coach_id</u> INT, <u>first_name</u> VARCHAR(50), <u>last_name</u> VARCHAR(50), <u>nationality</u> VARCHAR(50), <u>former_player_id</u> INT	<u>coach_id</u>	<u>former_player_id</u> → Player(<u>player_id</u>)	Each coach can reference a valid former player record. This supports the assumption that all coaches were formerly professional players.
Player Coach History	Stores the history of coaching relationships between players and coaches.	<u>history_id</u> INT, <u>player_id</u> INT, <u>coach_id</u> INT, <u>start_date</u> DATE, <u>end_date</u> DATE	<u>history_id</u>	<u>player_id</u> → Player(<u>player_id</u>); <u>coach_id</u> → Coach(<u>coach_id</u>)	Records when a coach started and ended coaching a player. <u>end_date</u> should be greater than or equal to <u>start_date</u> . This allows a player to have different coaches over time.

Figure 2.2.1.a - Login to the university remote lab environment using Parallels Client.

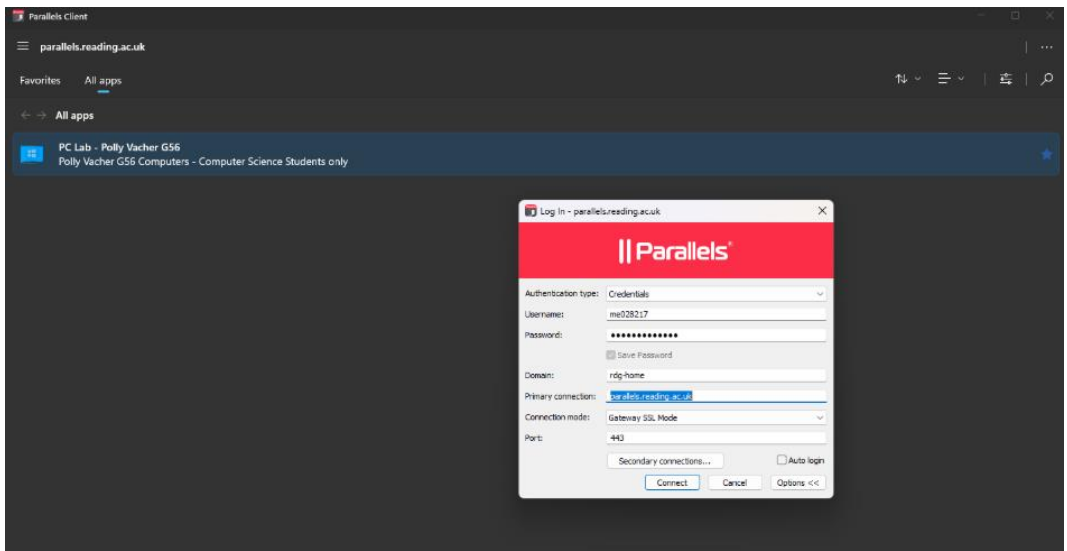
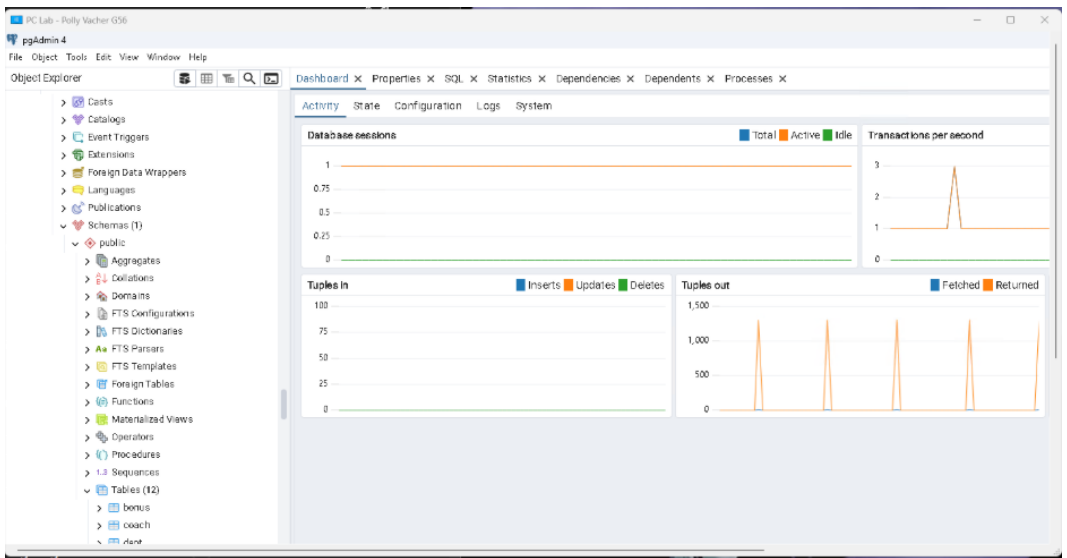


Figure 2.2.1.b - pgAdmin 4 environment used to connect to PostgreSQL and manage the tennis database.



SECTION 2.2

Figure 2.2.3.a - Player table structure/output

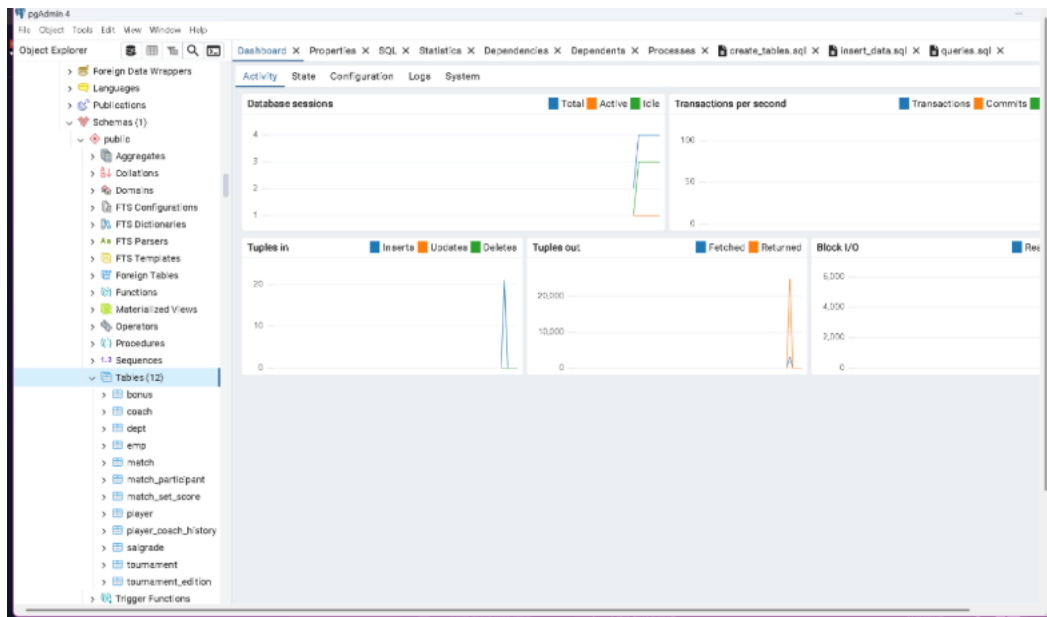


Figure 2.2.3.b - Coach table structure/output

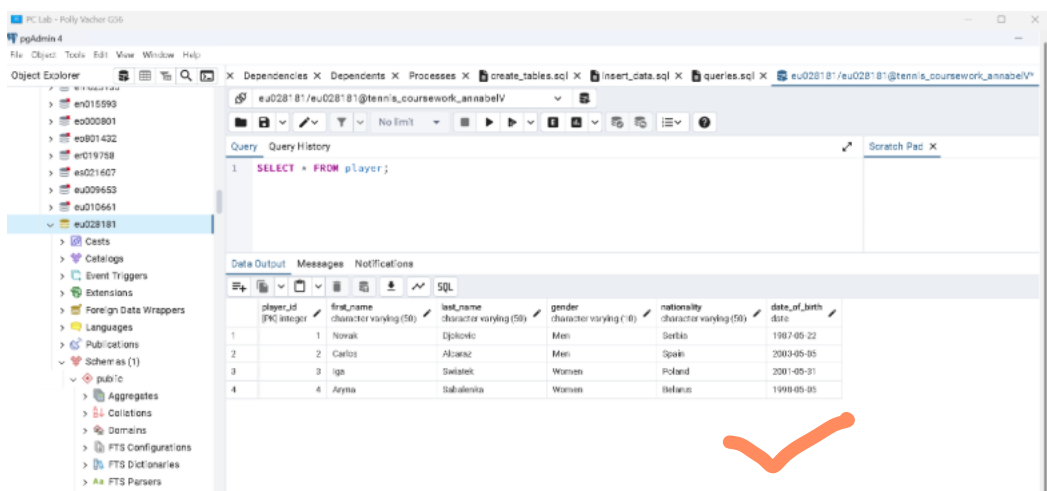


Figure 2.2.3.c - Tournament table structure/output

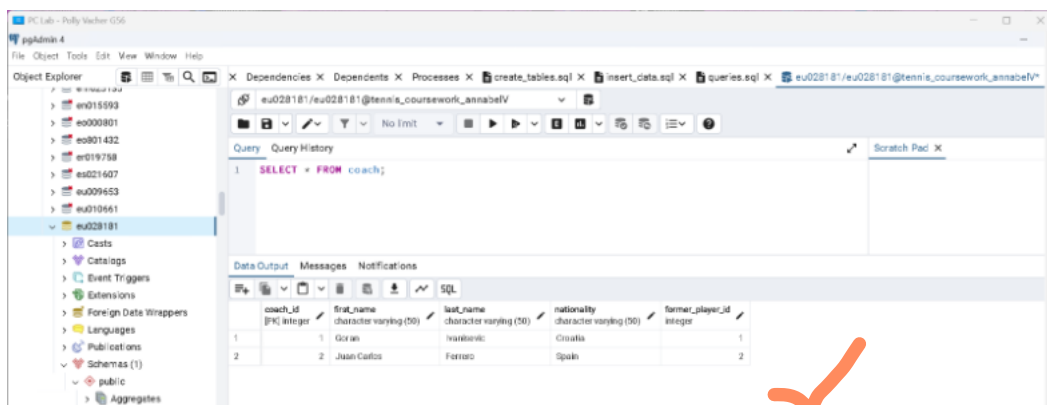


Figure 2.2.3.d - Tournament_Edition table structure/output

Query: `SELECT * FROM tournament;`

tournament_id	tournament_name	category	location	surface_type	is_grand_slam
1	Wimbledon	Grand Slam	London	Grass	true
2	Australian Open	Grand Slam	Melbourne	Hard	true

Figure 2.2.3.e - Match table structure/output

Query: `SELECT * FROM match;`

match_id	edition_id	round_name	match_type	gender_category	match_date	winner_side
1	1	Final	Singles	Men	2023-07-14	1
2	2	Final	Singles	Men	2024-07-28	2

Figure 2.2.3.f - Match_Participant table structure/output

Query: `SELECT * FROM match_participant;`

match_id	player_id	side_no
1	1	1
2	1	2
3	2	1
4	2	2

Figure 2.2.3.g - player_Coach_History / Match_Set_Score structure/output

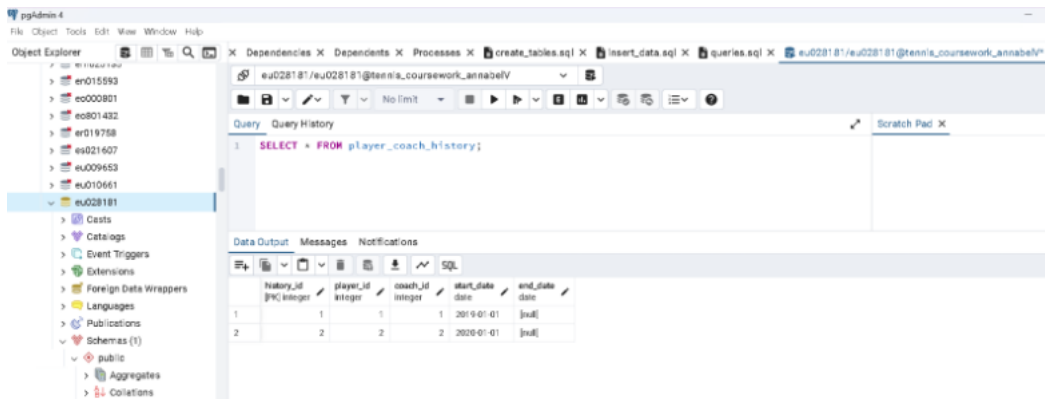


Figure 2.2.4.a - Example SQL-DML INSERT INTO statements used to populate the database tables.

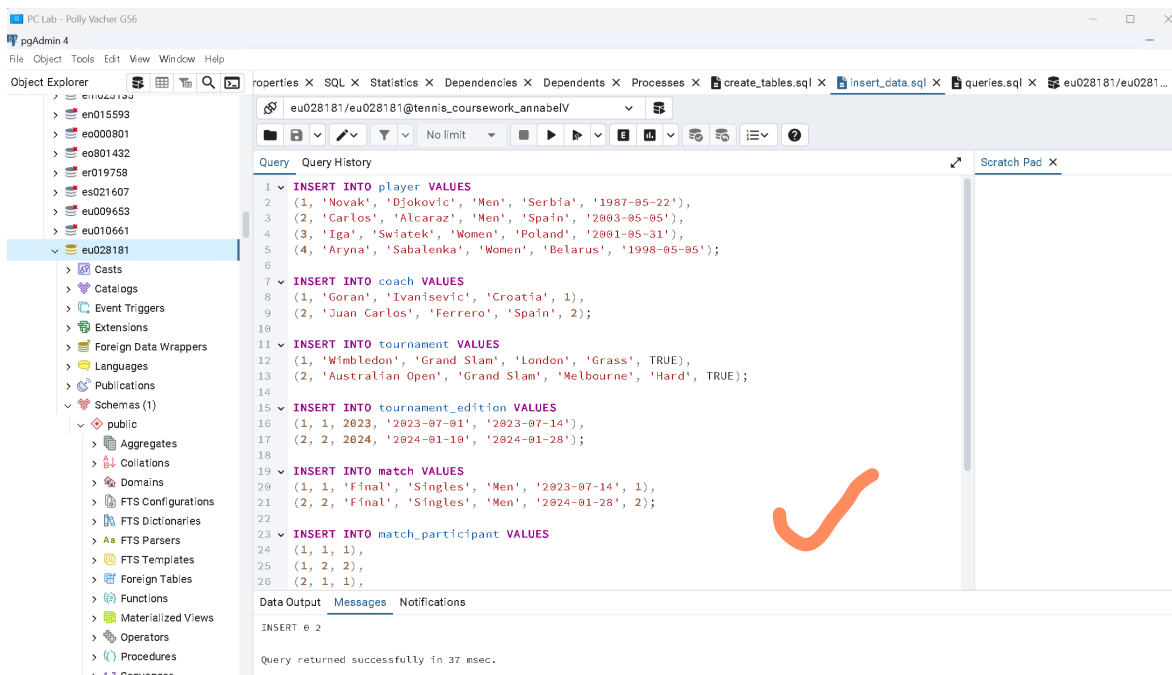


Figure 2.2.5.a - Record-count validation showing that key tables contain inserted records.

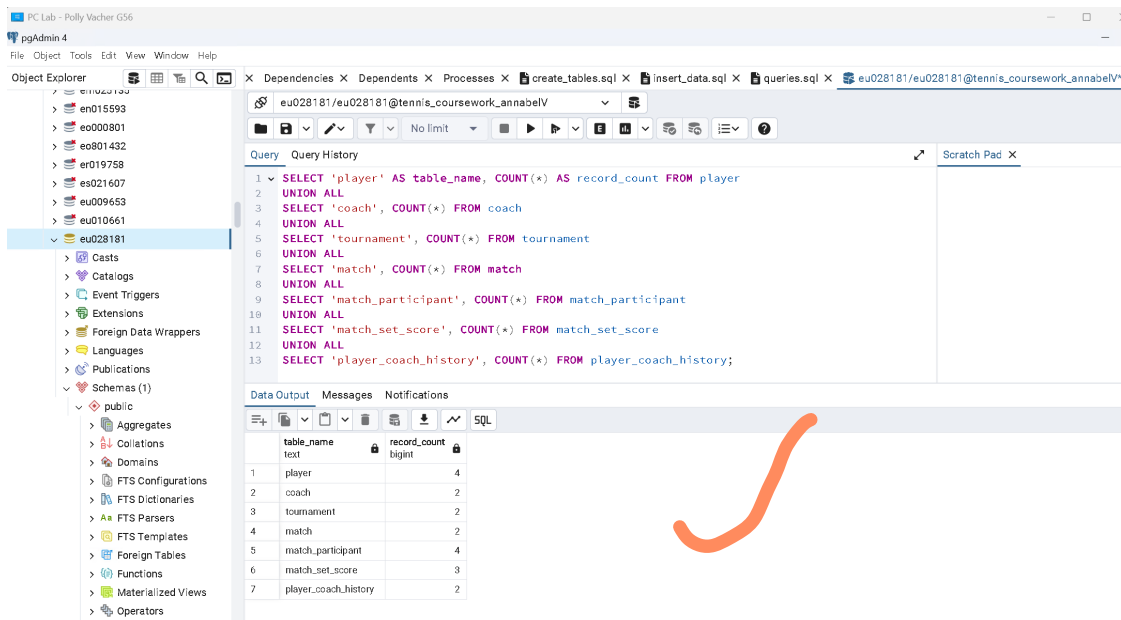


Figure 2.2.6.a - pgAdmin table list showing all 8 tables created.

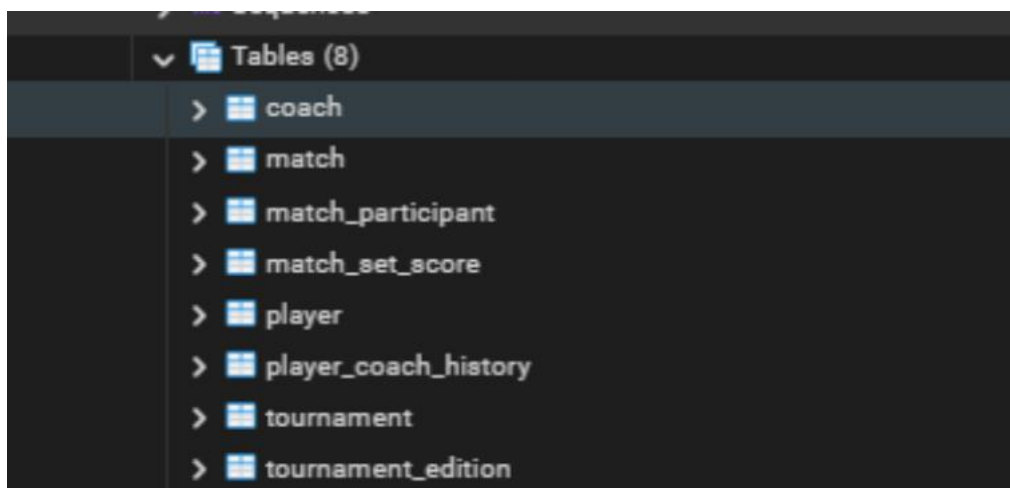


figure 2.2.6.b - CREATE TABLE SQL running successfully.

```
create_tables.sql X
tennis_db/postgres@PostgreSQL 18
Query Query History
1 CREATE TABLE player (
2   player_id INT PRIMARY KEY,
3   first_name VARCHAR(50) NOT NULL,
4   last_name VARCHAR(50) NOT NULL,
5   gender VARCHAR(10) NOT NULL CHECK (gender IN ('M', 'F')),
6   nationality VARCHAR(50) NOT NULL,
7   date_of_birth DATE NOT NULL
8 );
9
10 CREATE TABLE coach (
11   coach_id INT PRIMARY KEY,
12   first_name VARCHAR(50) NOT NULL,
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 249 msec.



Figure 2.2.6.c - table structure/columns/keys/constraints expanded in pgAdmin.

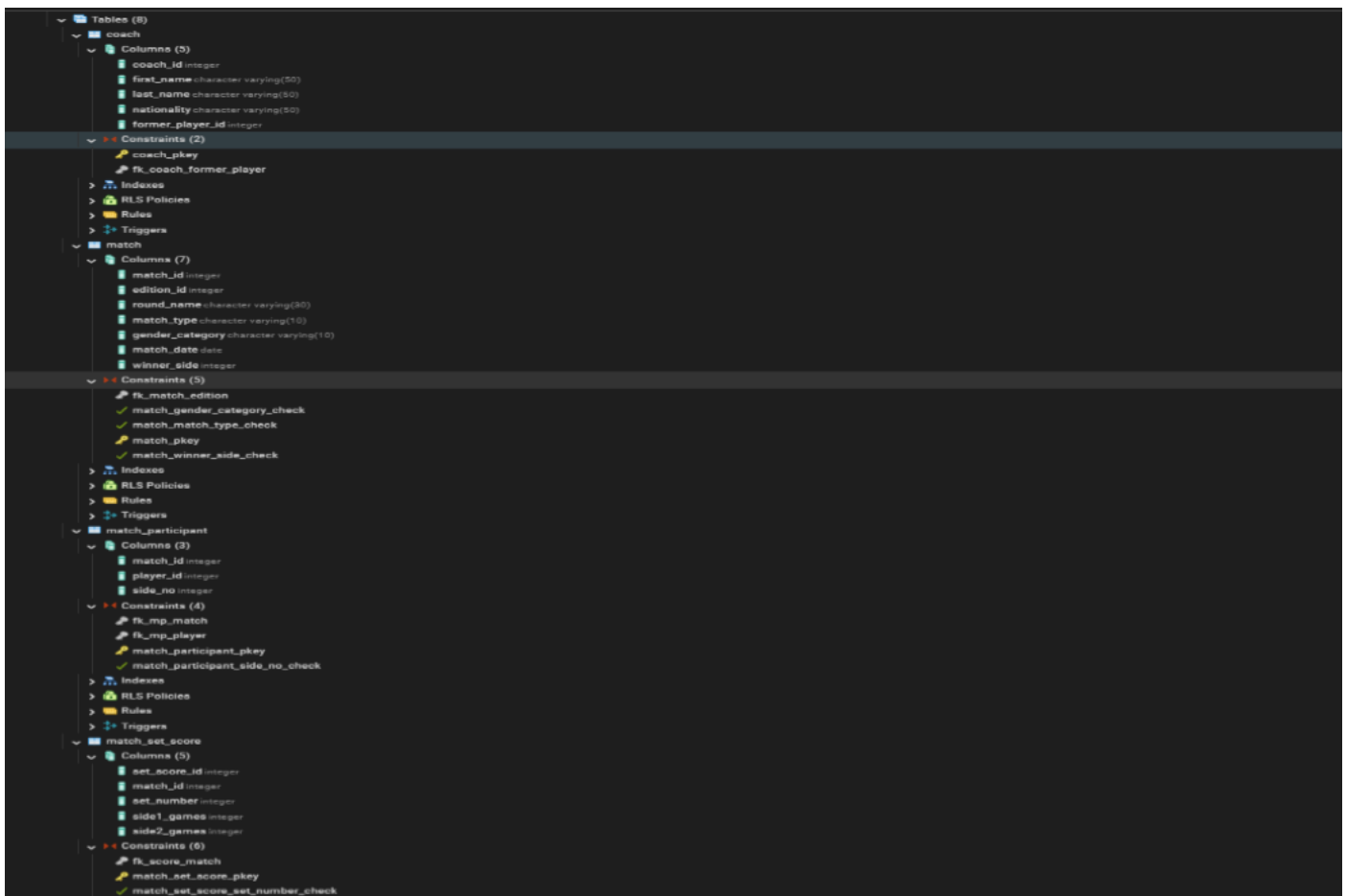


Figure 2.2.6.d - query checking match counts by tournament.

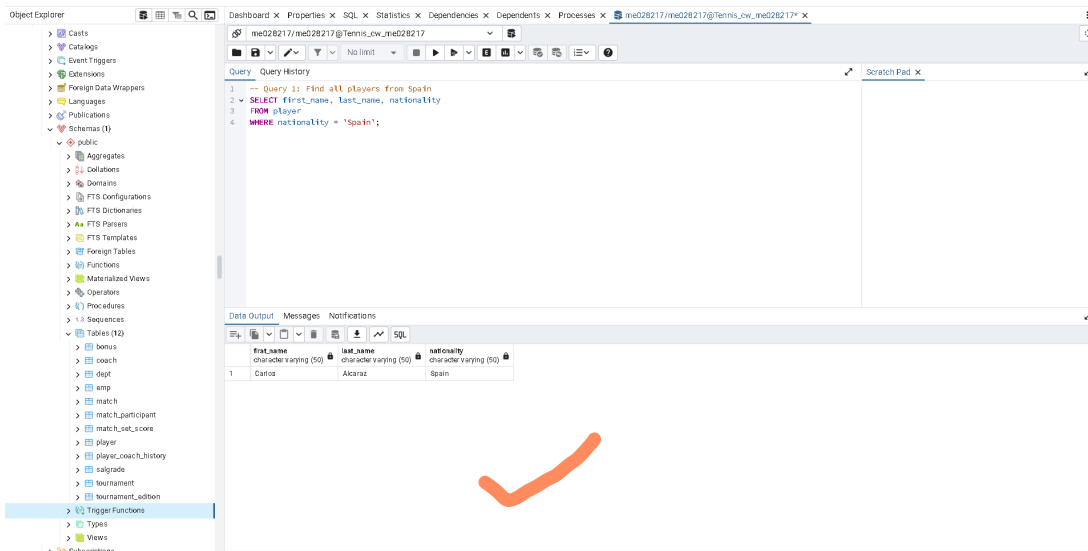


Figure 2.3.1.b - Simple grouping query using COUNT, GROUP BY, and ORDER BY to summarise players by nationality.

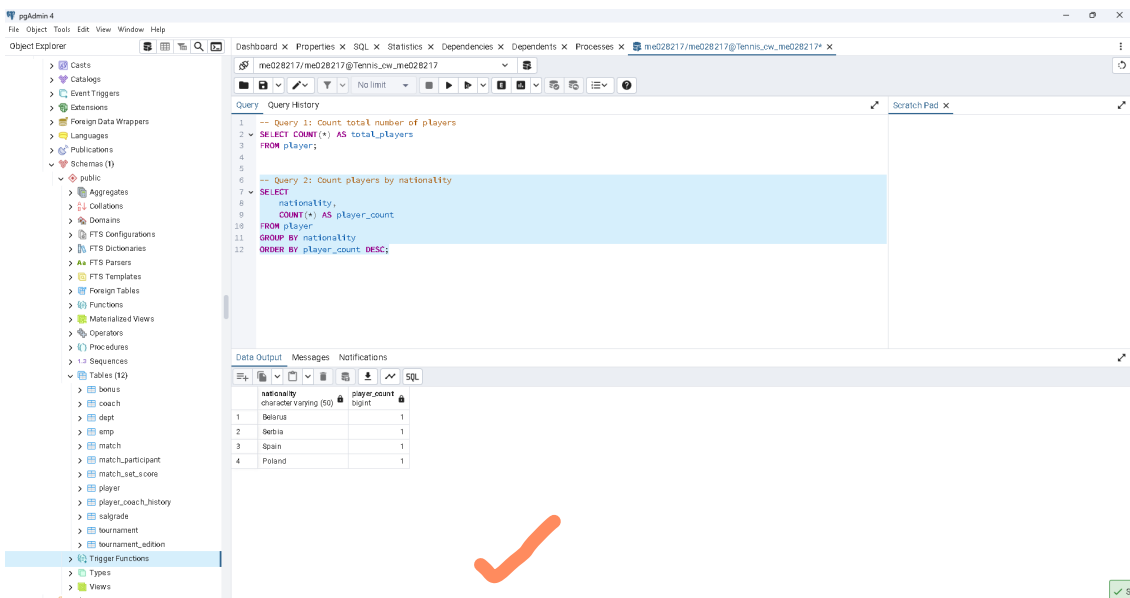


Figure 2.3.2.a - Complex query calculating match wins by player using joins between match, match_participant, and player.

The screenshot shows the pgAdmin 4 interface with a query window open. The query is a complex multi-table query that joins several tables to retrieve match scorelines. The query is as follows:

```

20 JOIN tournament_edition te
21 ON t.tournament_id = te.tournament_id
22 JOIN match m
23 ON te.edition_id = m.edition_id
24 GROUP BY t.tournament_name;
25
26
27 -- Query 4: Count wins by player
28 SELECT
29   p.first_name,
30   p.last_name,
31   COUNT(*) AS match_wins
32 FROM match m
33 JOIN match_participant mp
34 ON m.match_id = mp.match_id
35 JOIN player p
36 ON mp.player_id = p.player_id
37 WHERE mp.side_no = m.winner_side
38 GROUP BY p.player_id, p.first_name, p.last_name
39 ORDER BY match_wins DESC;

```

The output of the query is displayed in the Data Output tab, showing two rows of results:

first_name	last_name	match_wins
Novak	Djokovic	1
Carlos	Alcaraz	1

A red checkmark is visible on the right side of the query window, indicating a successful execution.

Figure 2.3.2.b - Complex multi-table query retrieving full match scorelines using match, tournament_edition, tournament, and match_set_score.

The screenshot shows the pgAdmin 4 interface with a query window open. The query is a complex multi-table query that joins several tables to retrieve match winners. The query is as follows:

```

1 -- Query 1: Show match winners using winner_side
2 SELECT
3   m.match_id,
4   t.tournament_name,
5   m.round_name,
6   p.first_name,
7   p.last_name,
8   m.match_date
9 FROM match m
10 JOIN match_participant mp
11 ON m.match_id = mp.match_id
12 JOIN player p
13 ON mp.player_id = p.player_id
14 JOIN tournament_edition te
15 ON m.edition_id = te.edition_id
16 JOIN tournament t
17 ON te.tournament_id = t.tournament_id
18 WHERE mp.side_no = m.winner_side;

```

The output of the query is displayed in the Data Output tab, showing two rows of results:

match_id	tournament_name	round_name	first_name	last_name	match_date
1	Tennis	Round 1	Novak	Djokovic	2023-09-01
2	Tennis	Round 1	Carlos	Alcaraz	2023-09-01

A red checkmark is visible on the right side of the query window, indicating a successful execution.

Additional Query Evidence: Further SQL query outputs used to test filtering, joins, aggregation and validation of the populated tennis database.

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Cast
- Catalogs
- Event Triggers
- Extensions
- Foreign Data Wrappers
- Languages
- Publications
- Schemas (1)
- public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (12)
 - bonus
 - coach
 - dept
 - emp
 - match
 - match_participant
 - match_set_score
 - player
 - player_coach_history

Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processes X me028217/me028217@Tennis_cw_me028217*

me028217/me028217@Tennis_cw_me028217

Query Query History

```
-- Query 1: Find all players from Spain
2 SELECT first_name, last_name, nationality
3 FROM player
4 WHERE nationality = 'Spain';
5
6
-- Query 2: List all female players
7 SELECT first_name, last_name, gender
8 FROM player
9 WHERE gender = 'Women';
10
```

Data Output Messages Notifications

first_name	last_name	gender
Iga	Swiatek	Women
Aryna	Sabalenka	Women



pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Cast
- Catalogs
- Event Triggers
- Extensions
- Foreign Data Wrappers
- Languages
- Publications
- Schemas (1)
- public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (12)
 - bonus
 - coach
 - dept
 - emp
 - match
 - match_participant
 - match_set_score
 - player
 - player_coach_history

Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processes X me028217/me028217@Tennis_cw_me028217*

me028217/me028217@Tennis_cw_me028217

Query Query History

```
-- Query 1: Find all players from Spain
2 SELECT first_name, last_name, nationality
3 FROM player
4 WHERE nationality = 'Spain';
5
6
-- Query 2: List all female players
7 SELECT first_name, last_name, gender
8 FROM player
9 WHERE gender = 'Women';
10
11
-- Query 3: Show all tournaments played on grass courts
12 SELECT tournament_name, surface_type
13 FROM tournament
14 WHERE surface_type = 'Grass';
15
16
```

Data Output Messages Notifications

tournament_name	surface_type
Wimbledon	Grass



pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Cast
- Catalogs
- Event Triggers
- Extensions
- Foreign Data Wrappers
- Languages
- Publications
- Schemas (1)
- public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (12)
 - bonus
 - coach
 - dept
 - emp
 - match
 - match_participant
 - match_set_score
 - player
 - player_coach_history

Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processes X me028217/me028217@Tennis_cw_me028217*

me028217/me028217@Tennis_cw_me028217

Query Query History

```
-- Query 4: Show players and their coaches
2 SELECT
3 p.first_name,
4 p.last_name,
5 c.first_name AS coach_first_name,
6 c.last_name AS coach_last_name
7 FROM player p
8 JOIN coach c ON p.player_id = c.former_player_id;-- Query 4: Show players and their coaches
9
10 SELECT
11 p.first_name,
12 p.last_name,
13 c.first_name AS coach_first_name,
14 c.last_name AS coach_last_name
15 FROM player p
16 JOIN coach c ON p.player_id = c.former_player_id;
```

Data Output Messages Notifications

first_name	last_name	coach_first_name	coach_last_name
Novak	Djokovic	Goran	ivanisevic
Carlos	Alcaraz	Juan Carlos	Ferrero



pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Cast
- Catalogs
- Event Triggers
- Extensions
- Foreign Data Wrappers
- Languages
- Publications
- Schemas (1)
- public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (12)
 - bonus
 - coach
 - dept
 - emp
 - match
 - match_participant
 - match_set_score
 - player
 - player_coach_history

Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processes X me028217/me028217@Tennis_cw_me028217*

me028217/me028217@Tennis_cw_me028217

Query Query History

```
-- Query 5: Show matches with tournament names
2 SELECT
3 m.match_id,
4 t.tournament_name,
5 m.round_name,
6 m.match_date
7 FROM match m
8 JOIN tournament t ON m.tournament_id = t.tournament_id;
9
10 JOIN tournament t ON m.tournament_id = t.tournament_id;
```

Data Output Messages Notifications

match_id	tournament_name	round_name	match_date
1	Wimbledon	Final	2023-07-14
2	Australian Open	Final	2024-01-28



pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Cast
- Catalogs
- Event Triggers
- Extensions
- Foreign Data Wrappers
- Languages
- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (12)
 - bonus
 - coach
 - dept
 - emp
 - match
 - match_participant

Dashboard Properties SQL Statistics Dependencies Dependents Processes me028217/me028217@Tennis_cw_me028217*

Query Query History

```
-- Query 1: Count total number of players
1 SELECT COUNT(*) AS total_players
2 FROM player;
```

Data Output Messages Notifications

total_players
4



pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Cast
- Catalogs
- Event Triggers
- Extensions
- Foreign Data Wrappers
- Languages
- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (12)
 - bonus
 - coach
 - dept
 - emp
 - match

Dashboard Properties SQL Statistics Dependencies Dependents Processes me028217/me028217@Tennis_cw_me028217*

Query Query History

```
-- Query 2: Count players by nationality
5 SELECT
6     nationality,
7     COUNT(*) AS player_count
8 FROM player
9 GROUP BY nationality
10 ORDER BY player_count DESC;
```

```
-- Query 3: Count matches by tournament
15 SELECT
16     tournament_name,
17     COUNT(m.match_id) AS total_matches
18 FROM tournament t
19 JOIN tournament_edition te
20 ON t.tournament_id = te.tournament_id
21 JOIN match m
22 ON te.edition_id = m.edition_id
23 GROUP BY t.tournament_name;
```

Data Output Messages Notifications

tournament_name	total_matches
Wimbledon	1
Australian Open	1



pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Cast
- Catalogs
- Event Triggers
- Extensions
- Foreign Data Wrappers
- Languages
- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (12)
 - bonus
 - coach
 - dept
 - emp
 - match
 - match_participant
 - match_set_score

Dashboard Properties SQL Statistics Dependencies Dependents Processes me028217/me028217@Tennis_cw_me028217*

Query Query History

```
18 WHERE sp.sides_no = m.winner_side;
19
20
21 -- Query 2: Show full match scoreline by set
22 SELECT
23     m.match_id,
24     t.tournament_name,
25     m.round_name,
26     m.match_date,
27     mss.set_number,
28     mss.sides1_games,
29     mss.sides2_games
30 FROM match m
31 JOIN tournament_edition te
32 ON m.edition_id = te.edition_id
33 JOIN tournament t
34 ON te.tournament_id = t.tournament_id
35 JOIN match_set_score mss
36 ON m.match_id = mss.match_id
37 ORDER BY m.match_id, mss.set_number;
```

Data Output Messages Notifications

match_id	tournament_name	round_name	match_date	set_number	sides1_games	sides2_games
1	Wimbledon	Final	2023-07-14	1	6	4
2	Wimbledon	Final	2023-07-14	2	6	7
3	Wimbledon	Final	2023-07-14	3	6	3



pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Cast
- Catalogs
- Event Triggers
- Extensions
- Foreign Data Wrappers
- Languages
- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (12)
 - bonus
 - coach
 - dept
 - emp
 - match
 - match_participant
 - match_set_score

Dashboard Properties SQL Statistics Dependencies Dependents Processes me028217/me028217@Tennis_cw_me028217*

Query Query History

```
39 mss.sides2_games
40 FROM match m
41 JOIN tournament_edition te
42 ON m.edition_id = te.edition_id
43 JOIN tournament t
44 ON te.tournament_id = t.tournament_id
45 JOIN match_set_score mss
46 ON m.match_id = mss.match_id
47 ORDER BY m.match_id, mss.set_number;
```

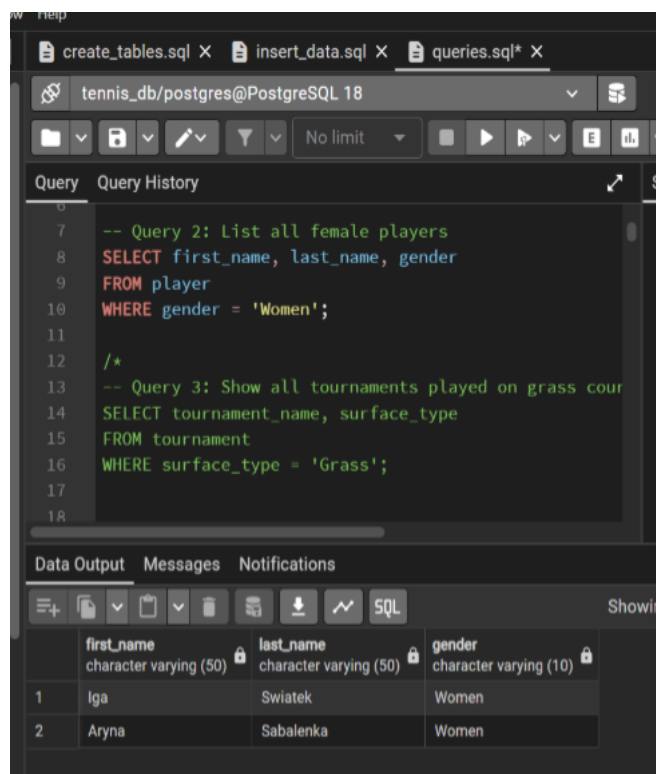
```
-- Query 3: find players who have both played matches and have coaching experience
48 SELECT DISTINCT
49     p.first_name,
50     p.last_name
51 FROM player p
52 JOIN match_participant mp
53 ON p.player_id = mp.player_id
54 JOIN coach c
55 ON c.player_id = c.former_player_id;
```

Data Output Messages Notifications

first_name	last_name
Carlos	Alcaraz
Novak	Djokovic



Figure 2.3.3.a - Simple query test retrieving all female players from the player table.



The screenshot shows a PostgreSQL query editor with the following SQL query:

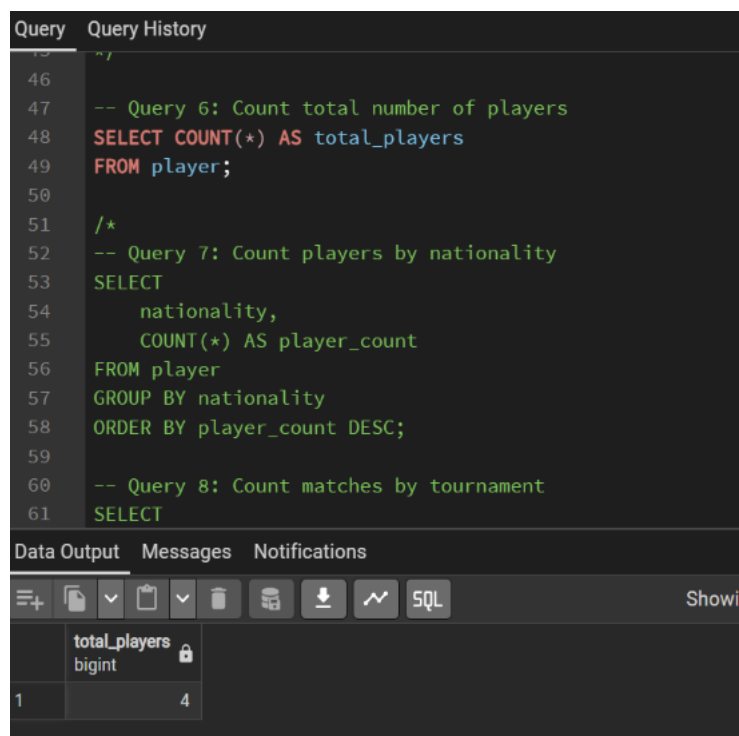
```
-- Query 2: List all female players
SELECT first_name, last_name, gender
FROM player
WHERE gender = 'Women';

/*
-- Query 3: Show all tournaments played on grass court
SELECT tournament_name, surface_type
FROM tournament
WHERE surface_type = 'Grass';
```

The results of the query are displayed in a table:

	first_name	last_name	gender
1	Iga	Swiatek	Women
2	Aryna	Sabalenka	Women

Figure 2.3.3.b - Simple query test using COUNT to validate the total number of player records.



The screenshot shows a PostgreSQL query editor with the following SQL query:

```
-- Query 6: Count total number of players
SELECT COUNT(*) AS total_players
FROM player;

/*
-- Query 7: Count players by nationality
SELECT
    nationality,
    COUNT(*) AS player_count
FROM player
GROUP BY nationality
ORDER BY player_count DESC;

-- Query 8: Count matches by tournament
SELECT
```

The results of the query are displayed in a table:

	total_players
1	4

Figure 2.3.3.c - complex query test retrieving full match scorelines using joins between match, tournament_edition, tournament and match_set_score.

Query Query History

```

87 -- Query 10: Show full match scoreline by set
88 SELECT
89     m.match_id,
90     t.tournament_name,
91     m.round_name,
92     m.match_date,
93     mss.set_number,
94     mss.side1_games,
95     mss.side2_games
96 FROM match m
97 JOIN tournament_edition te
98 ON m.edition_id = te.edition_id
99 JOIN tournament t
100 ON te.tournament_id = t.tournament_id
101 JOIN match_set_score mss
102 ON m.match_id = mss.match_id
103 ORDER BY m.match_id, mss.set_number;

```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1

match_id	tournament_name	round_name	match_date	set_number	side1_games	side2_games
integer	character varying (100)	character varying (30)	date	integer	integer	integer
1	1 Wimbledon	Final	2023-07-14	1	6	4
2	1 Wimbledon	Final	2023-07-14	2	5	7
3	1 Wimbledon	Final	2023-07-14	3	6	3



Figure 2.3.3.d - Complex query test calculating match wins by player using joins between match, match_participant and player.

Query Query History

```

71 */
72 -- Query 9: Count wins by player
73 SELECT
74     p.first_name,
75     p.last_name,
76     COUNT(*) AS match_wins
77 FROM match m
78 JOIN match_participant mp
79 ON m.match_id = mp.match_id
80 JOIN player p
81 ON mp.player_id = p.player_id
82 WHERE mp.side_no = m.winner_side
83 GROUP BY p.player_id, p.first_name, p.last_name
84 ORDER BY match_wins DESC;
85
86 /*
87 -- Query 10: Show full match scoreline by set

```

Data Output Messages Notifications

Showing r

first_name	last_name	match_wins
character varying (50)	character varying (50)	bigint
1 Novak	Djokovic	1
2 Carlos	Alcaraz	1

